

Classified as hazardous in accordance with the criteria of EPA New Zealand

## Section 1 - Identification

### Product Identifier

|                      |                            |
|----------------------|----------------------------|
| Product Name         | <u>1,2-Dichlorobenzene</u> |
| CAS No               | 95-50-1                    |
| Synonyms             | o-Dichlorobenzene          |
| Molecular Formula    | C6 H4 Cl2                  |
| Molecular Weight     | 147                        |
| Recommended Use      | Laboratory chemicals.      |
| Uses advised against | No Information available   |

|                         |                                                                                             |
|-------------------------|---------------------------------------------------------------------------------------------|
| Product Code            | 222050000; 222050025; 222055000                                                             |
| Address                 | Thermo Fisher Scientific New Zealand Ltd<br>244 Bush Road, Albany,<br>Auckland, New Zealand |
| Emergency Tel.          | CHEMTREC®<br>09 980 6780 or +64 9 980 6780                                                  |
| Telephone / Fax Numbers | Tel: 09 980 6700<br>Fax: 09 980 6788                                                        |
| E-mail address          | <a href="mailto:ANZinfo@thermofisher.com">ANZinfo@thermofisher.com</a>                      |

## Section 2 - Hazard(s) Identification

### Classification under Work Safe New Zealand

Classified as hazardous in accordance with the criteria of EPA New Zealand

HSNO Approval Number     HSR002509

### GHS Classification

#### Physical hazards

Flammable liquids

Category 4

#### Health hazards

Acute Oral Toxicity

Category 4

Acute Inhalation Toxicity - Vapors

Category 3

Skin Corrosion/Irritation

Category 2

Serious Eye Damage/Eye Irritation

Category 2

Skin Sensitization

Category 1

Germ Cell Mutagenicity

Category 2

Specific target organ toxicity - (single exposure)

Category 3

#### Environmental hazards

Acute aquatic toxicity  
Chronic aquatic toxicity

Category 1  
Category 1

**Label Elements****Signal Word****Danger****Hazard Statements**

H227 - Combustible liquid  
H302 - Harmful if swallowed  
H315 - Causes skin irritation  
H317 - May cause an allergic skin reaction  
H319 - Causes serious eye irritation  
H335 - May cause respiratory irritation  
H341 - Suspected of causing genetic defects if inhaled  
H410 - Very toxic to aquatic life with long lasting effects  
H331 - Toxic if inhaled

**Precautionary Statements****Prevention**

P210 - Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking  
P270 - Do not eat, drink or smoke when using this product  
P264 - Wash face, hands and any exposed skin thoroughly after handling  
P271 - Use only outdoors or in a well-ventilated area  
P272 - Contaminated work clothing should not be allowed out of the workplace  
P280 - Wear protective gloves/protective clothing/eye protection/face protection  
P273 - Avoid release to the environment

**Response**

P302 + P352 - IF ON SKIN: Wash with plenty of soap and water  
P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing  
P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
P312 - Call a POISON CENTER or doctor if you feel unwell  
P330 - Rinse mouth  
P370 + P378 - In case of fire: Use dry sand, dry chemical or alcohol-resistant foam to extinguish  
P362 + P364 - Take off contaminated clothing and wash it before reuse  
P391 - Collect spillage

**Storage**

P403 + P233 - Store in a well-ventilated place. Keep container tightly closed  
P405 - Store locked up

**Disposal**

P501 - Dispose of contents/ container to an approved waste disposal plant

**Other hazards which do not result in classification**

Toxic to terrestrial vertebrates

## Section 3 - Composition and Information on Ingredients

| Component         | CAS No  | Weight % |
|-------------------|---------|----------|
| o-Dichlorobenzene | 95-50-1 | >95      |

## Section 4 - First Aid Measures

### Description of first aid measures

|                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|--------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>General Advice</b>                      | If symptoms persist, call a physician.                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>New Zealand Emergency Tel.</b>          | CHEMTREC®<br>09 980 6780 or +64 9 980 6780                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Inhalation</b>                          | Remove to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur.                                                                                                                                                                                                                                                                                                                                                   |
| <b>Eye Contact</b>                         | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.                                                                                                                                                                                                                                                                                                                                                |
| <b>Skin Contact</b>                        | Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician.                                                                                                                                                                                                                                                                                                                                              |
| <b>Ingestion</b>                           | Clean mouth with water and drink afterwards plenty of water.                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Self-Protection of the First Aider</b>  | Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.                                                                                                                                                                                                                                                                                                               |
| <b>First Aid Facilities</b>                | Eyewash, safety shower and washroom.                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Most important symptoms and effects</b> | None reasonably foreseeable. May cause allergic skin reaction. Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting: Symptoms of allergic reaction may include rash, itching, swelling, trouble breathing, tingling of the hands and feet, dizziness, lightheadedness, chest pain, muscle pain or flushing: Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting |
| <b>Notes to Physician</b>                  | Treat symptomatically. Symptoms may be delayed.                                                                                                                                                                                                                                                                                                                                                                                                                |

## Section 5 - Fire Fighting Measures

### Suitable Extinguishing Media

Water spray, carbon dioxide (CO<sub>2</sub>), dry chemical, alcohol-resistant foam. Water mist may be used to cool closed containers.

### Extinguishing media which must not be used for safety reasons

No information available.

### Specific Hazards Arising from the Chemical

Combustible material. Containers may explode when heated. Keep product and empty container away from heat and sources of ignition. Thermal decomposition can lead to release of irritating gases and vapors. Do not allow run-off from fire-fighting to enter drains or water courses.

### Hazardous Combustion Products

Carbon monoxide (CO), Carbon dioxide (CO<sub>2</sub>), Hydrogen chloride gas.

### Special protective equipment and precautions for fire fighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

## Section 6 - Accidental Release Measures

### Personal Precautions, Protective Equipment and Emergency Procedures

#### Emergency procedures

Use personal protective equipment as required. Ensure adequate ventilation. Remove all sources of ignition. Take precautionary

measures against static discharges.

**Environmental Precautions**

Do not flush into surface water or sanitary sewer system. Do not allow material to contaminate ground water system. Prevent product from entering drains. Local authorities should be advised if significant spillages cannot be contained.

**Methods for Containment and Clean Up**

Soak up with inert absorbent material. Keep in suitable, closed containers for disposal. Remove all sources of ignition.

**Precautions to prevent secondary hazards**

Clean contaminated objects and areas thoroughly observing environmental regulations

**Reference to Other Sections**

Refer to protective measures listed in Sections 8 and 13.

## Section 7 - Handling and Storage

**Precautions for Safe Handling****Advice on safe handling**

Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Ensure adequate ventilation. Avoid ingestion and inhalation. Keep away from open flames, hot surfaces and sources of ignition.

**Hygiene Measures**

Handle in accordance with good industrial hygiene and safety practice. Keep away from food, drink and animal feeding stuffs. Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Wash hands before breaks and after work.

**Conditions for Safe Storage, Including any Incompatibilities****Storage Conditions**

Keep containers tightly closed in a dry, cool and well-ventilated place. Keep away from heat, sparks and flame.

**Incompatible Materials**

Strong oxidizing agents. Metals.

AS/NZS 2243.10:2004, Safety in laboratories - Storage of chemicals

AS 1940-2004 - The storage and handling of flammable and combustible liquids

## Section 8 - Exposure Controls and Personal Protection

**Control parameters****Exposure limits**

**NZ** - Workplace Exposure Standards and Biological Exposure Indices (6th edition). New Zealand Department of Labor

**UK** - EH40/2005 Work Exposure Limits, Fourth edition. Published 2020.

**AUS** - Exposure Standards for Atmospheric Contaminants in the Occupational Environment - Guidance Note on the Interpretation of Exposure Standards for Atmospheric Contaminants in the Occupational Environment [NOHSC:3008(1995)]

Adopted National Exposure Standards for Atmospheric Contaminants in the Occupational Environment [NOHSC:1003(1995)] updated in August, 2005. Safe Work Australia

**ACGIH** - Threshold Limit Values - Ceiling (TLV-C) guidelines by the American Conference of Governmental Industrial Hygienists (ACGIH) for controlling worker exposure to airborne chemical concentrations in the workplace.

| Component         | New Zealand WEL                                                                         | Australia                                                                                | ACGIH TLV                   | The United Kingdom                                                                                                       |
|-------------------|-----------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|-----------------------------|--------------------------------------------------------------------------------------------------------------------------|
| o-Dichlorobenzene | TWA: 10 ppm<br>TWA: 61 mg/m <sup>3</sup><br>STEL: 20 ppm<br>STEL: 122 mg/m <sup>3</sup> | STEL: 50 ppm<br>STEL: 301 mg/m <sup>3</sup><br>TWA: 25 ppm<br>TWA: 150 mg/m <sup>3</sup> | TWA: 25 ppm<br>STEL: 50 ppm | STEL: 50 ppm 15 min<br>STEL: 306 mg/m <sup>3</sup> 15 min<br>TWA: 25 ppm 8 hr<br>TWA: 153 mg/m <sup>3</sup> 8 hr<br>Skin |

**Biological limit values**

This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific

regulatory bodies

### Appropriate engineering controls

#### Engineering Measures

Ensure that eyewash stations and safety showers are close to the workstation location. Use explosion-proof electrical/ventilating/lighting equipment. Use only under a chemical fume hood. Ensure adequate ventilation, especially in confined areas. Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

### Individual protection measures, such as personal protective equipment

**Eye Protection** Goggles (Australian/New Zealand Standard AS/NZS 1337 - Eye protectors for Industrial applications)

**Hand Protection** Protective gloves

| Glove material | Breakthrough time | Glove thickness | AUS/NZ Standard | Glove comments                                                                 |
|----------------|-------------------|-----------------|-----------------|--------------------------------------------------------------------------------|
| Viton (R).     | > 480 minutes     | 0.7 mm          | AS/NZS 2161     | As tested under EN374-3 Determination of Resistance to Permeation by Chemicals |

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

**Skin and body protection** Long sleeved clothing

**Respiratory Protection** Use an AS/NZS 1716 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced. To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained in line with AS/NZS 1715 on the use and maintenance of respiratory protective devices

**Recommended Filter type:** Organic gases and vapours filter Type A Brown conforming to EN14387 (or AUS/NZ equivalent)

**Recommended half mask:-** Valve filtering: EN405 or Half mask: EN140 plus filter, EN 141 (or AUS/NZ equivalent)

**Hygiene Measures** Handle in accordance with good industrial hygiene and safety practice.

**Environmental exposure controls** Prevent product from entering drains. Do not allow material to contaminate ground water system. Local authorities should be advised if significant spillages cannot be contained.

## Section 9 - Physical and Chemical Properties

### Information on basic physical and chemical properties

|                                 |                               |                       |
|---------------------------------|-------------------------------|-----------------------|
| <b>Physical State</b>           | Liquid                        |                       |
| <b>Appearance</b>               | Clear                         |                       |
| <b>Odor</b>                     | No information available      |                       |
| <b>Odor Threshold</b>           | No data available             |                       |
| <b>pH</b>                       | No information available      |                       |
| <b>Melting Point/Range</b>      | -15 °C / 5 °F                 |                       |
| <b>Softening Point</b>          | No data available             |                       |
| <b>Boiling Point/Range</b>      | 179 - 180 °C / 354.2 - 356 °F |                       |
| <b>Flammability (liquid)</b>    | Combustible liquid            | On basis of test data |
| <b>Flammability (solid,gas)</b> | Not applicable                | Liquid                |
| <b>Explosion Limits</b>         | <b>Lower</b> 2.2 Vol%         |                       |
|                                 | <b>Upper</b> 12 Vol%          |                       |

|                                         |                          |                          |
|-----------------------------------------|--------------------------|--------------------------|
| Flash Point                             | 67 °C / 152.6 °F         | Method - CC (closed cup) |
| Autoignition Temperature                | 640 °C / 1184 °F         |                          |
| Decomposition Temperature               | No data available        |                          |
| Viscosity                               | No data available        |                          |
| Water Solubility                        | 0.13 g/l                 |                          |
| Solubility in other solvents            | No information available |                          |
| Partition Coefficient (n-octanol/water) |                          |                          |
| Component                               | log Pow                  |                          |
| o-Dichlorobenzene                       | 3.433                    |                          |
| Vapor Pressure                          | 1.3 mbar @ 20 °C         |                          |
| Density / Specific Gravity              | 1.3 g/cm3 @20°C          |                          |
| Bulk Density                            | Not applicable           | Liquid                   |
| Vapor Density                           | No data available        | (Air = 1.0)              |
| Particle characteristics                | Not applicable (liquid)  |                          |

**Other information**

|                      |                                        |
|----------------------|----------------------------------------|
| Molecular Formula    | C6 H4 Cl2                              |
| Molecular Weight     | 147                                    |
| Explosive Properties | explosive air/vapour mixtures possible |

## Section 10 - Stability and Reactivity

|                                  |                                                                                                                   |
|----------------------------------|-------------------------------------------------------------------------------------------------------------------|
| Reactivity                       | None known, based on information available                                                                        |
| Stability                        | Stable under normal conditions.                                                                                   |
| Sensitivity to Mechanical Impact | No information available                                                                                          |
| Sensitivity to Static Discharge  | No information available                                                                                          |
| Hazardous Polymerization         | No information available.                                                                                         |
| Hazardous Reactions              | None under normal processing.                                                                                     |
| Conditions to Avoid              | Incompatible products, Heat, flames and sparks, Keep away from open flames, hot surfaces and sources of ignition. |
| Incompatible Materials           | Strong oxidizing agents, Metals.                                                                                  |
| Hazardous Decomposition Products | Carbon monoxide (CO). Carbon dioxide (CO <sub>2</sub> ). Hydrogen chloride gas.                                   |

## Section 11 - Toxicological Information

**Acute Effects****Information on likely routes of exposure****Product Information**

|            |                                                                                                       |
|------------|-------------------------------------------------------------------------------------------------------|
| Inhalation | Irritating to respiratory system. May be harmful if inhaled.                                          |
| Eyes       | Irritating to eyes.                                                                                   |
| Skin       | Irritating to skin. May be harmful in contact with skin.                                              |
| Ingestion  | Harmful if swallowed. Ingestion may cause gastrointestinal irritation, nausea, vomiting and diarrhea. |

**Numerical measures of toxicity**

|                     |                                                                  |
|---------------------|------------------------------------------------------------------|
| (a) acute toxicity; |                                                                  |
| Oral                | Category 4                                                       |
| Dermal              | Based on available data, the classification criteria are not met |

## Inhalation

Category 4

| Component         | LD50 Oral                 | LD50 Dermal               | LC50 Inhalation     |
|-------------------|---------------------------|---------------------------|---------------------|
| o-Dichlorobenzene | LD50 = 1516 mg/kg ( Rat ) | LD50 > 10 g/kg ( Rabbit ) | 14,04 mg/L/4h (Rat) |

(b) skin corrosion/irritation; Category 2  
 Test method OECD 404  
 Test species rabbit  
 Observational endpoint Erythema/Eschar = = 1.56  
 Oedema = = 1

(c) serious eye damage/irritation; Category 2  
 Test method OECD 405  
 Test species rabbit  
 Observation end point Iris lesion = 0.06  
 Cornea opacity = 0  
 Redness of the conjunctivae = 0.6  
 Oedema of the conjunctivae = 0.11

(d) respiratory or skin sensitization;  
 Respiratory Based on available data, the classification criteria are not met  
 Skin Category 1

| Component                            | Test method                                       | Test species | Study result |
|--------------------------------------|---------------------------------------------------|--------------|--------------|
| o-Dichlorobenzene<br>95-50-1 ( >95 ) | OECD Test Guideline 429 Local<br>Lymph Node Assay | mouse        | Sensitizer   |

## Sensitization

May cause sensitization by skin contact

(e) germ cell mutagenicity; Based on available data, the classification criteria are not met

| Component                            | Test method                                                | Test species                 | Study result |
|--------------------------------------|------------------------------------------------------------|------------------------------|--------------|
| o-Dichlorobenzene<br>95-50-1 ( >95 ) | OECD Test Guideline 476<br>Gene cell mutation              | in vitro<br>Animal germ cell | Positive     |
|                                      | OECD Test Guideline 471<br>Bacterial Reverse Mutation Test | in vitro<br>Bacteria         | negative     |
|                                      | OECD Test Guideline 473<br>Chromosomal aberration assay    | in vitro<br>Animal germ cell | negative     |
|                                      | OECD Test Guideline 474<br>Mouse micronucleus assay        | in vivo<br>Animal germ cell  | negative     |

(f) carcinogenicity; Based on available data, the classification criteria are not met  
 There are no known carcinogenic chemicals in this product

(g) reproductive toxicity; Based on available data, the classification criteria are not met

(h) STOT-single exposure; Category 3

Results / Target organs Respiratory system

(i) STOT-repeated exposure; Based on available data, the classification criteria are not met

Test method Chronic Toxicity  
 Test species / Duration Rat / 90 days

**Study result**  
**Route of exposure**  
**Target Organs**

NOAEL = 125 mg/kg  
 Oral  
 None known.

**(j) aspiration hazard;** Based on available data, the classification criteria are not met

**Other Adverse Effects** Tumorigenic effects have been reported in experimental animals.

#### Symptoms / effects, both acute and delayed

Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting. Symptoms of allergic reaction may include rash, itching, swelling, trouble breathing, tingling of the hands and feet, dizziness, lightheadedness, chest pain, muscle pain or flushing. Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting.

## Section 12 - Ecological Information

### Ecotoxicity

**Aquatic ecotoxicity** Very toxic to aquatic organisms, may cause long-term adverse effects in the aquatic environment. The product contains following substances which are hazardous for the environment.

| Component         | Freshwater Fish                                                                                                                                                                                                                                                                                                                                                        | Water Flea                                    | Freshwater Algae                                                                                                                                                                           | Microtox                                                                     |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------|
| o-Dichlorobenzene | LC50: 4.8 - 6.6 mg/L, 96h static (Lepomis macrochirus)<br>LC50: = 5.2 mg/L, 96h flow-through (Brachydanio rerio)<br>LC50: 42.6 - 80.4 mg/L, 96h static (Pimephales promelas)<br>LC50: 8.23 - 10.9 mg/L, 96h flow-through (Pimephales promelas)<br>LC50: 1.44 - 1.73 mg/L, 96h flow-through (Oncorhynchus mykiss)<br>LC50: = 5.8 mg/L, 96h static (Pimephales promelas) | EC50: = 0.74 mg/L, 48h Static (Daphnia magna) | EC50: = 91.6 mg/L, 96h (Pseudokirchneriella subcapitata)<br>EC50: 61.2 - 181 mg/L, 72h (Pseudokirchneriella subcapitata)<br>EC50: = 2.2 mg/L, 96h static (Pseudokirchneriella subcapitata) | EC50 = 4.76 mg/L 5 min<br>EC50 = 4.98 mg/L 15 min<br>EC50 = 5.99 mg/L 30 min |

**Terrestrial ecotoxicity** There is no data for this product

**Persistence and Degradability** Not readily biodegradable

**Persistence** May persist, based on information available.

| Component                          | Degradability       |
|------------------------------------|---------------------|
| o-Dichlorobenzene<br>95-50-1 (>95) | 0 % (28d) OECD 301C |

**Degradation in sewage treatment plant** Contains substances known to be hazardous to the environment or not degradable in waste water treatment plants.

**Bioaccumulative Potential** May have some potential to bioaccumulate

| Component         | log Pow | Bioconcentration factor (BCF) |
|-------------------|---------|-------------------------------|
| o-Dichlorobenzene | 3.433   | 90 - 260 dimensionless        |

**Mobility** The product is insoluble and sinks in water. The product evaporates slowly. Spillage unlikely to penetrate soil. . Is not likely mobile in the environment due its low water solubility. Spillage unlikely to penetrate soil



**Other adverse effects**

**Endocrine Disruptor Information**  
**Persistent Organic Pollutant**  
**Ozone Depletion Potential**

This product does not contain any known or suspected endocrine disruptors  
This product does not contain any known or suspected substance  
This product does not contain any known or suspected substance

## Section 13 - Disposal Considerations

**Waste treatment methods**

**Waste from Residues/Unused Products**

Do not allow into drains or watercourses or dispose of where ground or surface waters may be affected. Wastes, including emptied containers, are controlled wastes and should be disposed of in accordance with all federal, E.P.A., state and local regulations. Assure conformity with all applicable regulations.

**Contaminated Packaging**

Dispose of this container to hazardous or special waste collection point.

**Other Information**

Disposal agencies or waste contractors must comply with the New Zealand Hazardous Substances (Disposal) Regulations. Do not flush to sewer. Waste codes should be assigned by the user based on the application for which the product was used. Do not empty into drains. Do not let this chemical enter the environment.

## Section 14 - Transport Information

| Component                            | Hazchem Code |
|--------------------------------------|--------------|
| o-Dichlorobenzene<br>95-50-1 ( >95 ) | 2Z           |

**NZS 5433:2020**

**UN-No** UN1591  
**Proper Shipping Name** o-DICHLOROBENZENE  
**Hazard Class** 6.1  
**Packing Group** III

**IATA**

**UN-No** UN1591  
**Proper Shipping Name** o-DICHLOROBENZENE  
**Hazard Class** 6.1  
**Packing Group** III

**IMDG/IMO**

**UN-No** UN1591  
**Proper Shipping Name** O-DICHLOROBENZENE  
**Hazard Class** 6.1  
**Packing Group** III

**Environmental hazards**

Dangerous for the environment  
Product is a marine pollutant according to the criteria set by IMDG/IMO

**Transport in bulk according to Annex II of MARPOL 73/78 and the IBC Code**

Not applicable, packaged goods

**Special Precautions**

No special precautions required. Please refer to the applicable dangerous goods regulations for additional information.

Additional information None known

## Section 15 - Regulatory Information

### Safety, health and environmental regulations/legislation specific for the substance or mixture

|                      |           |
|----------------------|-----------|
| HSNO Approval Number | HSR002509 |
|----------------------|-----------|

#### National Regulations

There are no applicable tolerable exposure limits or environmental exposure limits according to the EPA Controls for Hazardous Substances

#### Certified handlers, tracking and controlled substance license requirements

Certified handlers are required for some substances. This includes substances requiring a controlled substance license, and most explosives, vertebrates toxic agents, and certain fumigants. Acutely toxic substances which are a Category 1 or 2, such as pesticides also require Certified handlers. Please check the Health and Safety at Work Act 2015 for further information. Tracking is required for some highly hazardous substances. These substances need to be under the control of an appropriately trained person or appropriately secured. Please check the Health and Safety at Work Act 2015 for further information.

#### Prohibition or notification/licensing requirements

Shown below are details of specific prohibition/notifications or licencing requirements when they apply.

#### International Regulations

**Ozone Depletion Potential** This product does not contain any known or suspected substance

**Persistent Organic Pollutant** This product does not contain any known or suspected substance

**Rotterdam Convention (PIC)** Not applicable

#### Authorisation/Restrictions according to EU REACH

| Component         | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|-------------------|---------------------------------------------------------------------|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| o-Dichlorobenzene | -                                                                   | Use restricted. See item 75. (see link for restriction details)               | -                                                                                                     |

<https://echa.europa.eu/substances-restricted-under-reach>

#### International Inventories

New Zealand (NZIoC), Australia (AICS), Europe (EINECS/ELINCS/NLP), Korea (KECL), China (IECSC), Taiwan (TCSI), Japan (ENCS), Japan (ISHL), Canada (DSL/NDL), Philippines (PICCS). US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

| Component         | CAS No  | NZIoC | AICS | EINECS    | ELINCS | NLP | KECL     | IECSC | TCSI |
|-------------------|---------|-------|------|-----------|--------|-----|----------|-------|------|
| o-Dichlorobenzene | 95-50-1 | X     | X    | 202-425-9 | -      | -   | KE-10066 | X     | X    |

| Component         | CAS No  | TSCA | TSCA Inventory notification - Active-Inactive | DSL | NDL | PICCS | ISHL | ENCS |
|-------------------|---------|------|-----------------------------------------------|-----|-----|-------|------|------|
| o-Dichlorobenzene | 95-50-1 | X    | ACTIVE                                        | X   | -   | X     | X    | X    |

Legend: X - Listed '-' - Not Listed

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

## Section 16 - Other Information

**This safety data sheet complies with the requirements of the EPA Hazardous Substances (Hazard Classification) Notice 2020 and WorkSafe New Zealand Regulations**

**Legend**

|                                                                                                      |                                                                                                                              |
|------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| <b>NZIoC</b> - New Zealand Inventory of Chemicals                                                    | <b>AICS</b> - Australian Inventory of Chemical Substances                                                                    |
| <b>TSCA</b> - United States Toxic Substances Control Act Section 8(b) Inventory                      | <b>EINECS/ELINCS</b> - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances |
| <b>DSL/NDL</b> - Canadian Domestic Substances List/Non-Domestic Substances List                      | <b>ENCS</b> - Japanese Existing and New Chemical Substances                                                                  |
| <b>IECSC</b> - Chinese Inventory of Existing Chemical Substances                                     | <b>KECL</b> - Korean Existing and Evaluated Chemical Substances                                                              |
| <b>PICCS</b> - Philippines Inventory of Chemicals and Chemical Substances                            | <b>CAS</b> - Chemical Abstracts Service                                                                                      |
| <b>TWA</b> - Time Weighted Average                                                                   | <b>ACGIH</b> - American Conference of Governmental Industrial Hygienists                                                     |
| <b>IARC</b> - International Agency for Research on Cancer                                            | <b>PNEC</b> - Predicted No Effect Concentration                                                                              |
| <b>NZS 5433:2020</b> - Transport of Dangerous Goods on Land                                          | <b>OECD</b> - Organisation for Economic Co-operation and Development                                                         |
| <b>ICAO/IATA</b> - International Civil Aviation Organization/International Air Transport Association | <b>IMO/IMDG</b> - International Maritime Organization/International Maritime Dangerous Goods Code                            |
| <b>MARPOL</b> - International Convention for the Prevention of Pollution from Ships                  | <b>ADG</b> - Australian Code for the Transport of Dangerous Goods by Road and Rail                                           |
| <b>LD50</b> - Lethal Dose 50%                                                                        | <b>LC50</b> - Lethal Concentration 50%                                                                                       |
| <b>EC50</b> - Effective Concentration 50%                                                            | <b>ATE</b> - Acute Toxicity Estimate                                                                                         |
| <b>WEL</b> - Workplace Exposure Limit                                                                | <b>RPE</b> - Respiratory Protective Equipment                                                                                |
| <b>DNEL</b> - Derived No Effect Level                                                                | <b>NOEC</b> - No Observed Effect Concentration                                                                               |
| <b>POW</b> - Partition coefficient Octanol:Water                                                     | <b>BCF</b> - Bioconcentration factor                                                                                         |
| <b>vPvB</b> - very Persistent, very Bioaccumulative                                                  | <b>PBT</b> - Persistent, Bioaccumulative, Toxic                                                                              |
| <b>VOC</b> - (Volatile Organic Compound)                                                             |                                                                                                                              |

**Key literature references and sources for data**

HSNO classifications provided in the New Zealand Chemical Classification Information Database (CCID).

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadviser - LOLI, Merck index, RTECS

EPA Guide to classifying hazardous substances in New Zealand

EPA - Assigning a product to an existing HSNO approval guide

**Training Advice**

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

First aid for chemical exposure, including the use of eye wash and safety showers.

Chemical incident response training.

|                         |                |
|-------------------------|----------------|
| <b>Revision Date</b>    | 10-Mar-2023    |
| <b>Revision Summary</b> | Not applicable |

**Disclaimer**

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**End of Safety Data Sheet**